Understanding the Impact of Kernel Choice on Support Vector Machine Performance

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GitHub - https://github.com/Ssairamreddy06/MachineLearning_SVM.git

Introduction :- Machine learning is often used for classification problems, where the aim is to classify data points into different classes based on given features. One popular method for classification is known as Support Vector Machines (SVMs). SVMs are used to create an optimal decision boundary that maximizes the margin between different classes. However, many real-world problems are nonlinear and can't be separated linearly.

For such problems, SVMs use something called a kernel function. The kernel function helps to create nonlinear decision boundaries. This is because many problems can be separated linearly in higher dimensions. This is known as the kernel trick. Here, we are going to discuss different types of kernel functions and how they can be used for different problems. We are going to use a dataset that is created with the scikit-learn library and compare different types of kernel functions.

Methodology :-

Dataset Generation

A two-class dataset was generated using the make_moons dataset generator provided by scikit-learn. This dataset was chosen due to its non-linearly separable nature, which makes it suitable for testing kernel-based SVM models.

- Samples: 400
- Noise level: 0.25 (this value was chosen to introduce some realistic noise into the data)
- Random state: 42 (this value was chosen to ensure reproducibility)

The dataset was then split into training and testing sets using stratified sampling.

Data Preprocessing

The reason for the sensitivity of the Support Vector Machine to feature scaling led to the embedding of each model within a scikit-learn Pipeline, which includes the following:

1. StandardScaler

- Scaling the features to zero mean and unit variance
- Facilitate a fair comparison of the results across kernels

2. SVC classifier

- The kernel changes depending on the experiment

Model Selection

The following SVM models were considered:

1. Linear Kernel SVM

- Baseline model
- Determines if dataset is separable by a line

2.Polynomial Kernel SVM

- moderate complex non-linear data

3.RBF Kernel SVM

- Highly flexible
- Should do best on curved surfaces

All models used the same regularization parameter **C = 1.0** and default gamma="scale" where applicable.

Kernel Functions

Linear Kernel

- Assumes data is linearly separable.
- Produces a straight-line decision boundary.
- Fastest and simplest kernel.

Polynomial Kernel

- Captures non-linear interactions up to degree d
- More flexible than linear, less flexible than RBF
- Can model curved boundaries

RBF (Radial Basis Function) Kernel

- Extremely flexible
- Creates smooth, highly non-linear decision boundaries
- Controlled by parameter

Results :-

Kernel	Train Accuracy	Test Accuracy
Linear	0.818	0.892
Polynomial	0.807	0.867

RBF	0.921	0.942

Figure 1 :- Linear Kernel SVM Decision Boundary with Training and Test Accuracy.

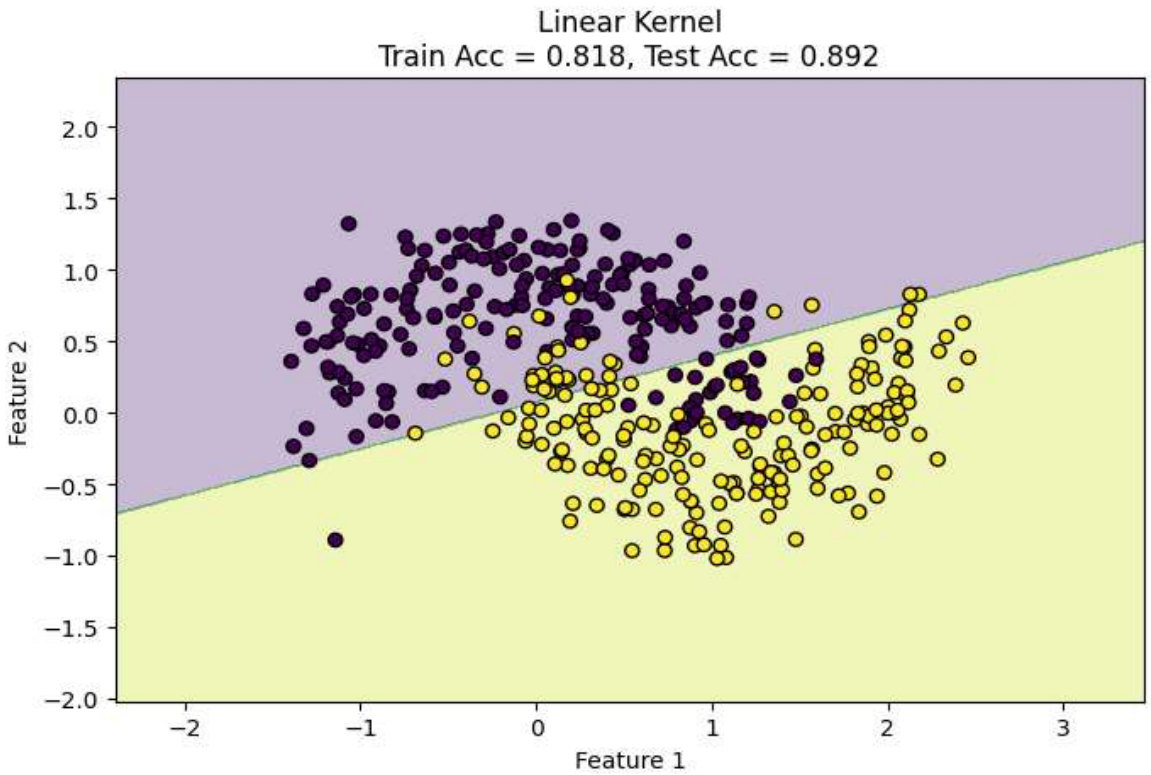


Figure 2 :- Decision Boundary of SVM with Polynomial Kernel (degree = 3).

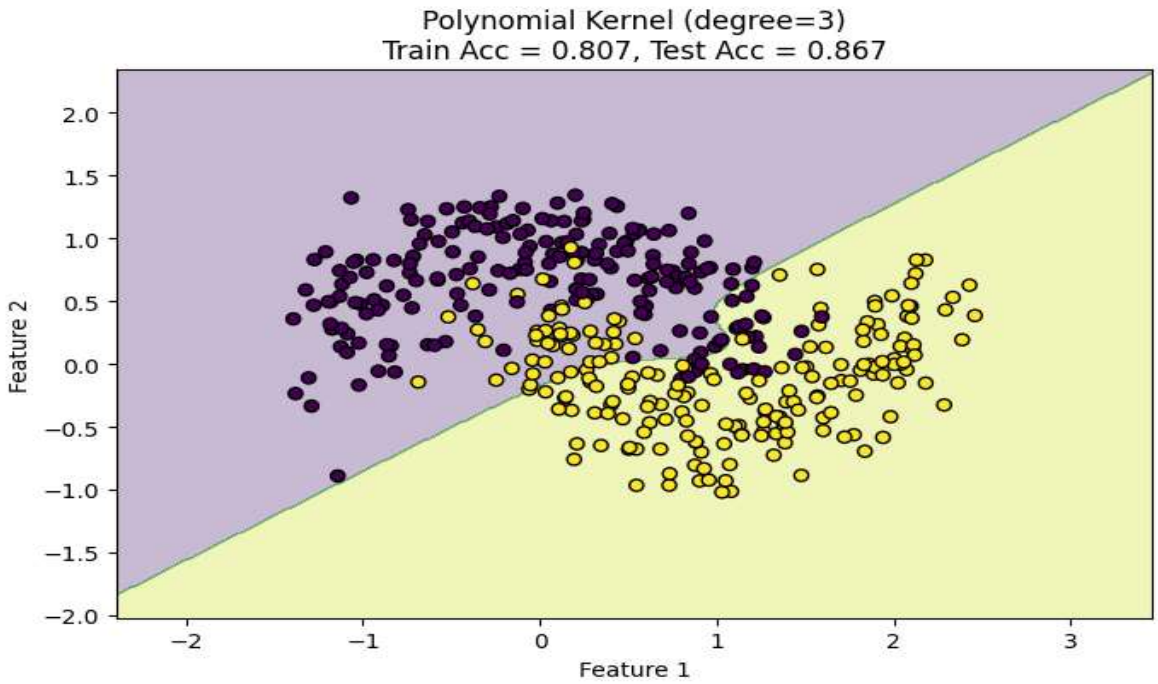


Figure 3 :- Decision Boundary of SVM with RBF Kernel (Train Acc = 0.921, Test Acc = 0.942).

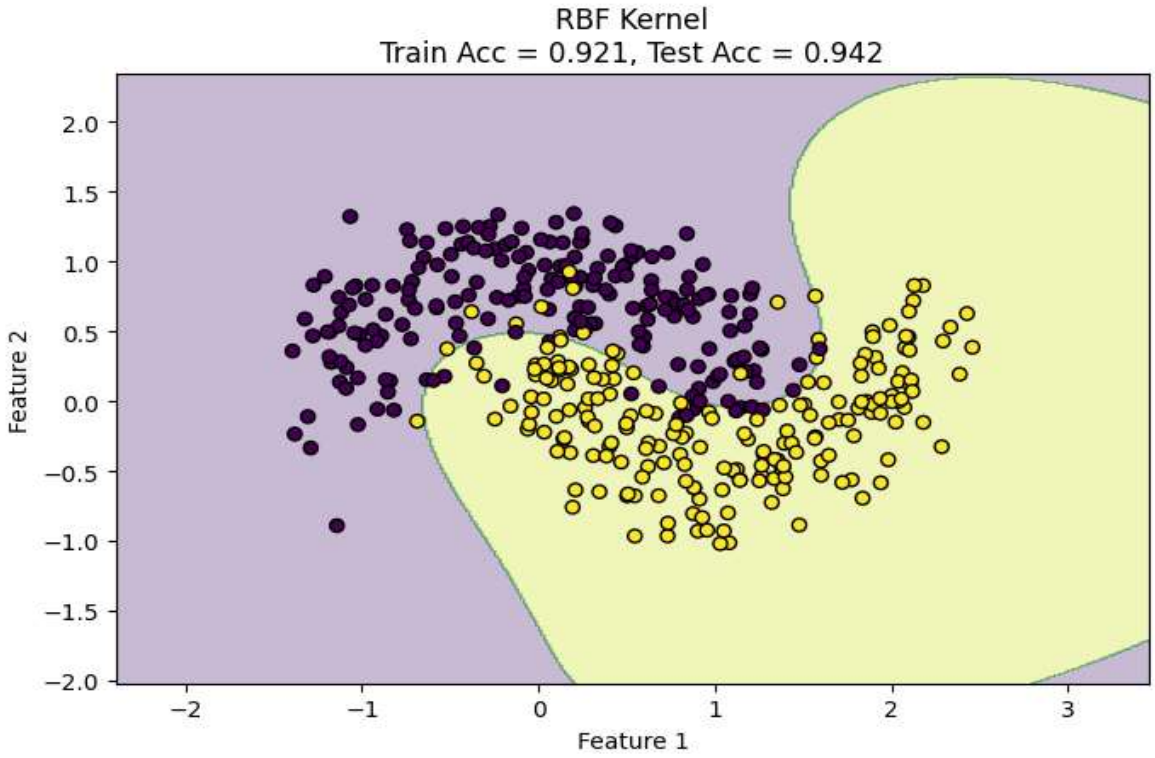
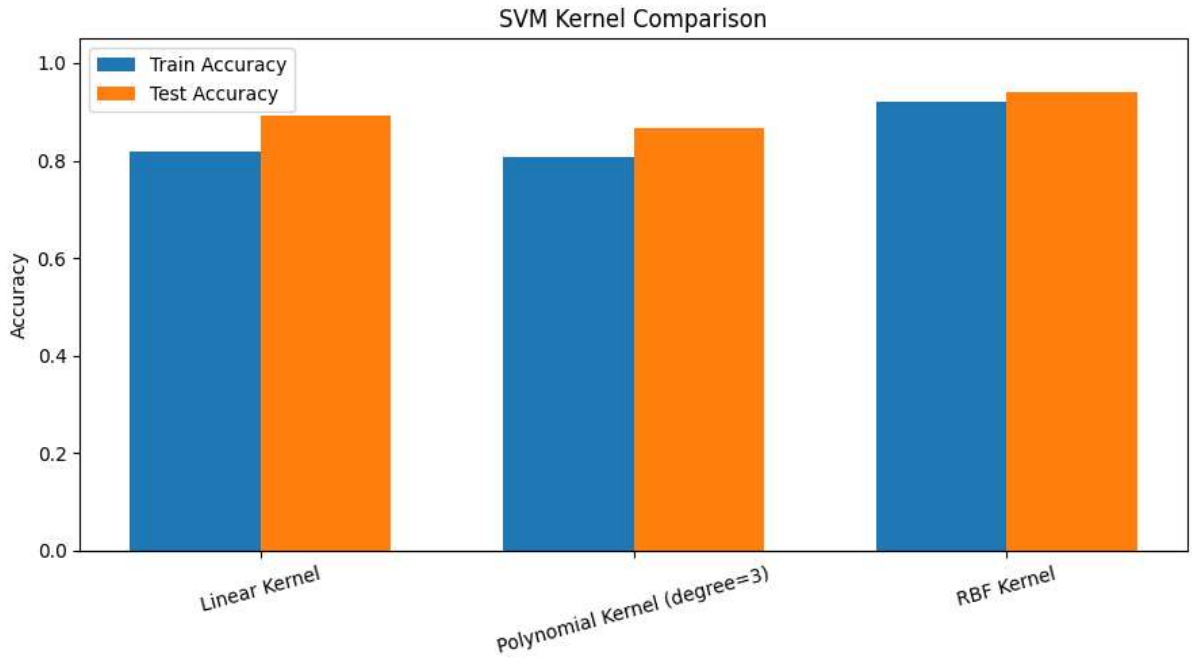


Figure 4 :- Comparison of Training and Test Accuracy Across SVM Kernels.



The Linear kernel uses a decision boundary that is essentially a straight line. The Linear kernel worked fairly well on this dataset because it was able to achieve an accuracy of 0.818 in training and an accuracy of 0.892 in testing; however, it is not effective in dealing with the curved nature of the data in this moon-shaped.

The Polynomial kernel of degree 3 was also used to incorporate curved boundary conditions. It has more ability to learn complex patterns but in this case has slightly lower accuracy with 0.807 in training and 0.867 in testing.

The most powerful kernel for non-linear data is the RBF kernel, as it creates smooth and flexible boundaries, and it is also the best kernel used in the classification problem, as its train accuracy was 0.921 and test accuracy was 0.942.

Discussion :-

Why did the RBF kernel perform best?

The reason why the RBF kernel performed the best is that it is specifically developed to work well with non-linear and complex decision boundaries, which are exactly what are found in the curved structure of the make_moons dataset. Unlike linear and polynomial kernels, the RBF kernel uses distance to measure similarities. This makes the RBF kernel very effective when dealing with data sets whose classes are intertwined and have irregular structures, like your data set.

The RBF kernel is also locally adaptable. This means that the RBF kernel can change its boundary shape locally. This local adaptability of the RBF kernel makes it very effective at capturing local patterns without imposing a global shape on the data. This explains why the RBF kernel performed the best by achieving the highest accuracy of 0.921 on the train set and 0.942 on the test set.

Why did the Polynomial Kernel underperform?

The linear kernel performed remarkably well (0.818 train, 0.892 test), indicating that even though this is a non-linear dataset, there is some linear separability to be found. This is to say that a linear boundary is capable of capturing some aspect of this data, especially where one class is more clearly distinguishable from another.

One of the biggest drawbacks of the linear kernel is that it is not sufficiently malleable to capture the curved and entwined nature of these moon data points. It is unable to bend and conform to these local features and will therefore always misclassify points near these curved boundaries.

Insights from the Linear Kernel

The linear kernel performed remarkably well (0.818 train, 0.892 test), indicating that even though this is a non-linear dataset, there is some linear separability to be found.

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Overfitting Analysis

The small difference between the training set and the test set for all the kernels shows that there was minimal overfitting. For example, for the RBF kernel, there is a slight difference between the train set and the test set, where the train set has an accuracy of 0.921 and the test set has an accuracy of 0.942. This shows that the model has generalized well instead of memorizing the data. This is also shown by the classification report and the confusion matrix.

The results have also shown the bias-variance tradeoff. The linear kernel had high bias and low variance, resulting in underfitting. The polynomial kernel had moderate bias and variance. The RBF kernel had low bias by allowing high variance. This shows that the RBF kernel performed the best without overfitting.

Conclusion :- This study has shown that the kernel used has a significant impact in the performance of Support Vector Machine models. This is evident by comparing the performance of the linear, polynomial, and RBF kernels in a non-linear dataset. It is evident that each kernel has its own way of representing the relationship in the dataset. This has shown that the selection of the kernel is not arbitrary and should be carefully done in line with the nature of the dataset.

The results indicate that the RBF kernel has the best performance overall, with the highest accuracy in both training and tests. This is due to the RBF kernel's ability to handle highly non-linear data by creating smooth and flexible decision boundaries. This helps the RBF kernel generalize well, especially in the presence of noisy data. However, the linear kernel, although computationally efficient and easy to understand, lacks the flexibility required to handle the complex data set. This is due to the inability of the linear kernel to handle complex data sets. The polynomial kernel, although more flexible than the linear kernel, is data-dependent and has difficulties in the presence of noisy data and when the degree of the polynomial is not correctly chosen.

Moreover, the significance of hyperparameter tuning and evaluation techniques such as cross-validation is also emphasized. These are crucial to ensure not only the accuracy of the model but also its reliability. The marginal difference between the training accuracy and test accuracy for all the kernels shows low overfitting, thus ensuring the good regularisation of the model and the size of the data being sufficient for learning.

From a practical point of view, this study again highlights that there is no best kernel that can be applied to all machine learning problems. Rather, the choice of the best kernel must be based on the nature of the data and the complexity of the decision boundary that is to be identified. For problems where patterns are highly nonlinear, such as in medical diagnosis, anomaly detection, and image classification, using an RBF kernel can be very useful.

To conclude, the choice of the RBF kernel has been identified as the best option for this dataset based on its ability to be flexible and generalise effectively. However, the choice of the appropriate kernel depends on the problem. It is important to understand the advantages and limitations of each kernel in developing effective and reliable machine learning models.

References :-

1. Rasmussen, C. E., & Williams, C. K. I. (2006). *Gaussian Processes for Machine Learning*. MIT Press.
2. Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer.